

CS F364: Design & Analysis of Algorithms

Quiz 6 — June 16, 2026

Coverage: Greedy Algorithms

i

Note

Instructions:

Time: 15 minutes.

Write your answers clearly on paper, scan, and upload to Google Classroom.

Show all steps for full credit.

Question: Interval Scheduling — Multiple Rooms

You are scheduling the following intervals in a venue with multiple identical rooms. Every interval must be scheduled in some room, and no two intervals in the same room may overlap. You want to use the **minimum number of rooms**.

Interval	I_1	I_2	I_3	I_4	I_5	I_6
Start	1	2	3	5	6	7
Finish	5	4	6	7	8	9

Describe the greedy algorithm for this problem (sorting rule and assignment rule). Run it step-by-step on the given intervals, showing which room each interval is assigned to. State the total number of rooms needed and explain why it is optimal.